

Meeting Minutes 1

Project Name: Financial and Operational Analysis of U.S. Hospitals Using Hospital Provider Cost Report Data (2011–2022)

Team Roles & Responsibilities

Name	Role	Responsibility Summary
Manjunath Muthineni	Team Leader, ML Data Integration & Evaluation Lead	Leads planning, coordinates meetings, designs the ML workflow, plans safe dataset integration, and defines evaluation strategy.
Rajesh Thota	ML Documentation & Reporting Lead	Maintains documentation structure, ensures clarity, and aligns work with rubric.
Neha	ML Research & Domain Methodology Lead	Provides healthcare finance input and validates efficiency logic.
Abhishekh Choudhary	Data Quality & Feature Engineering Lead	Plans data cleaning, join validation, and feature preparation.
Krishna Chaitanya Venuturumilli	ML Support & Review Lead	Reviews assumptions, checks scope, and supports planning discussions.
Professor Manjari Maheshwari	Faculty Advisor	Guides academic rigor, scope, and methodology decisions.

Meeting Minutes 1A (In-Class – With Professor)

(Project Direction, Data Strategy & Front-End Expectations)

Date and Time

12 Jan 2026, 12:00 PM – 3:00 PM

Location

St. Clair College, One Riverside Drive, Room 1013

Attendees

- Professor Manjari Maheshwari
- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

- Neha (0873038)

Agenda

- Review previous semester data work and continuity
- Discuss data supplementation for stronger ML models
- Discuss expectations for front-end interface
- Clarify project scope and academic expectations

Discussion Points

- **Manjunath** explained the work done in the previous semester, including how hospital cost data was cleaned, checked, and prepared for analysis. He explained that keeping the same structure helps the team build ML models without introducing errors.
- **Professor Maheshwari** asked the group what steps they will take to improve ML performance beyond the existing dataset. She explained that in real projects, ML quality improves when more relevant data is added carefully.
- **Abhishekh** explained the risks of joining datasets, such as duplicate hospital records, mismatched IDs, and leakage. He emphasized that join rules must be defined early to avoid errors later.

- **Rajesh** highlighted that all decisions must be clearly written in the report so the professor can see the group's thinking process.
- **Krishna** discussed the front-end requirement and explained that it should only display results and insights, not act as a deployed system.
- **Professor Maheshwari** reminded the group to focus on fewer models but explain them well, instead of adding unnecessary complexity.

Decisions Made

- Use additional public datasets to improve feature quality.
- Create and document safe join rules before integration.
- Limit the front-end to showing ML results and explanations.
- Remain in planning phase until methodology is approved.

Action Items

- Shortlist public datasets for integration.
- Draft join and leakage control plan.
- Outline front-end design at a high level.
- Prepare questions for the next meeting.

Any concerns (if any)

- Risk of expanding scope too quickly without approval.

Next Meeting Date and Time

12 Jan 2026, 3:00 PM – 3:40 PM

Minutes Prepared By

Krishna Chaitanya Venuturumilli

Meeting Minutes 1B (Group Meeting – Separate)

(Methodology Translation & Role Alignment)

Date and Time

12 Jan 2026, 3:00 PM – 3:40 PM

Location

St. Clair College, One Riverside Drive, Room 1013

Attendees

- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Convert professor feedback into a clear plan
- Confirm roles and responsibilities
- Decide how the group will review work

Discussion Points

- The group discussed how to turn professor guidance into clear steps that everyone can follow.
- **Neha** explained why Cost-to-Charge Ratio (CCR) is a reliable measure of hospital efficiency and why it fits both prediction and classification.
- **Abhishekh** explained that features must adjust for hospital size so the model does not favor larger hospitals.
- **Rajesh** explained how documentation will follow each stage of the ML process.
- **Krishna** reminded the group to keep assumptions the same across all models.
- **Manjunath** proposed planning evaluation before building models to avoid weak results.

Decisions Made

- Roles were confirmed and agreed upon.
- Weekly group meetings were scheduled.
- No model building until planning is complete.

Action Items

- Draft evaluation plan.
- Draft documentation outline.
- Draft feature engineering assumptions.
- Draft data quality checklist.

Any concerns (if any)

- Keeping everyone aligned as tasks run in parallel.

Next Meeting Date and Time

13 Jan 2026, 8:00 AM – 10:00 AM

Minutes Prepared By

Rajesh Thota

Meeting Minutes 2 (In-Class – With Professor)

(ML Theory, GenAI Discussion & Learning from Other Groups)

Date and Time

13 Jan 2026, 8:00 AM – 10:00 AM

Location

St. Clair College, One Riverside Drive, Room 1004

Attendees

- Professor Manjari Maheshwari
- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Understand ML theory and model justification
- Learn about AI and Generative AI evolution
- Share ideas with other groups
- Discuss possible scope expansion

Discussion Points

- **Professor Maheshwari** explained how AI started with rule-based systems and moved toward ML and Generative AI.
- She explained how Data Science, ML, AI, and GenAI work together in modern projects.
- The professor encouraged the group to not limit the project to ML only and to use GenAI if it helps explain results.
- The group discussed safe GenAI uses, such as generating summaries of model outputs or explaining results in simple language.

- The class shared ideas with other groups, which helped the team better understand different approaches.
- The group agreed GenAI should be used only if it clearly adds value.

Decisions Made

- ML will remain the main method.
- GenAI will be considered optional.
- Any GenAI use must be documented clearly.

Action Items

- Research GenAI use cases.
- Update methodology to include optional GenAI.
- Prepare questions for next professor meeting.

Any concerns (if any)

- Risk of making the project too complex if GenAI is overused.

Next Meeting Date and Time

14 Jan 2026, 6:00 PM – 7:30 PM

Minutes Prepared By

Neha

Meeting Minutes 3 (Group Meeting – Separate)

(Dataset Integration, Feasibility & Scope Finalization)

Date and Time

14 Jan 2026, 6:00 PM – 7:30 PM

Location

Online (Microsoft Teams)

Attendees

- Manjunath Muthineni (0878795)
- Rajesh Thota (0873782)
- Neha (0873038)
- Abhishekh Choudhary (0873460)
- Krishna Chaitanya Venuturumilli (0874277)

Absent

None

Agenda

- Finalize dataset integration approach
- Identify safe join keys
- Discuss data limits and risks

Discussion Points

- **Manjunath** explained the final plan to use CMS HCRIS as the main dataset and connect other public datasets to it.
- **Abhishekh** explained how join checks and validation rules will be used to avoid duplication or missing records.
- **Neha** explained that wage data is not publicly available and suggested using staffing and expense data as substitutes.
- **Rajesh** explained why all data limits must be clearly written in the report.
- **Krishna** checked that assumptions remained the same across all datasets.
- The group agreed that clean, well-documented data is better than more data with higher risk.

Decisions Made

- Wage data excluded.
- Proxy variables approved.
- Limitations will be documented clearly.

Action Items

- Write dataset integration section.
- Write join validation rules.
- Write proxy justification.
- Write limitations section.

Any concerns (if any)

- Explaining limitations clearly during final defense.

Next Meeting Date and Time

20 Jan 2026 (In-class meeting with professor)

Minutes Prepared By

Abhishekh Choudhary